N162AV (S/N: KONUK10295948) — Modem Absent From USB Bus Diagnostic Report

Immfly Ground Operations / IT Session: October 11, 2026

Field	Value
Machine	**N162AV** (S/N: KONUK10295948)
Affected component	Sierra Wireless EM7565 cellular modem (USB hub port `1-4.1`)
Initial symptom	No cellular connectivity. ModemManager reported `state: failed` with `failed reason: sim-missing`; `ppp0` was DOWN.
USB identification	**ABSENT** — No Sierra Wireless device (`1199:9091` or `1199:9090`) on the bus. Hub port 1-4.1 shows no activity, while the other ports on the same hub enumerate normally.
Root cause	Not determined. The modem was present and in normal mode at the start of the session, but stopped enumerating after the unit was powered off to reseat the SIM, and did not return after a full power cut. Software cannot tell whether the modem, its USB connection or its power is at fault.
Comparison	Same signature as N120AV (modem absent from the USB bus). On N109AV a similar absence cleared after a reboot; on N162AV it did not clear after a full power cut.
Recommended action	**Physical inspection and likely replacement of the modem module.** Software-level recovery exhausted; no firmware flashing is possible while the modem is not on the bus.

NARRATIVE SUMMARY

N162AV had no cellular connectivity. At the start of the session the modem was visible to the system: the udev symlinks `/dev/modemAT` and `/dev/modemPPP` pointed to `ttyUSB4` and `ttyUSB5`, and `lsusb` showed it in normal operating mode (`1199:9091`). It enumerated late, about 149 seconds after boot. ModemManager reported the modem as `failed` with `sim-missing`, and a direct SIM status query with `qmicli` failed with `endpoint hangup`, so the SIM state could not be read.

Eight accumulated `pppd` processes were found and cleared, with the connectivity cron job paused meanwhile. A logical modem reset through ModemManager brought the modem back as a new device, but it stayed in `sim-missing`. The unit was then powered off to reseat the SIM.

After power-on, the modem did not return. `mmcli -L` reported no modems and `lsusb` showed no Sierra Wireless device, with 8 minutes of uptime, well beyond the roughly 2.5 minutes this unit normally needs. The USB topology showed the hub on port 4 enumerating normally, with the FTDI serial adapter on its port 2 and the Teensy display controller on its port 3, but nothing on port 1, where the modem sits. The kernel log contained no enumeration errors for that port (no `unable to enumerate` or `error -71`), meaning the port reported no device at all.

A full power cut was then performed (power disconnected for 30 seconds) together with a check of the modem's seating. The modem still did not enumerate. Because the SIM could not be read before the modem disappeared, and the modem is no longer on the bus, the original `sim-missing` symptom could not be investigated further, and it is not possible from software to tell whether the fault lies in the modem module, its USB connection or its power supply.

N162AV (S/N: KONUK10295948) — Troubleshooting Steps & Command Appendix

TROUBLESHOOTING STEPS PERFORMED

1. **Initial state check:** The udev symlinks matched the rules (`modemAT` to ttyUSB4, `modemPPP` to ttyUSB5). The modem attached at about 149 s after boot. `mmcli` showed `failed` / `sim-missing`, power state on, and `ppp0` DOWN.
2. **SIM status query:** `qmicli --uim-get-card-status` failed with `CID allocation failed ... endpoint hangup`. The SIM state could not be read.
3. **Safety check:** `lsusb` confirmed `1199:9091` (normal mode, not QDL) with ttyUSB0 to ttyUSB5 present and no errors in the kernel log.
4. **Process cleanup:** `pgrep -c pppd` returned 8 accumulated processes. The connectivity cron job was paused and all `pppd` processes were killed (0 remaining); nothing held ttyUSB5.
5. **Logical reset:** `mmcli -m 0 --reset` succeeded. After 45 s the modem reappeared as a new device and remained in `sim-missing`.
6. **Power-off for SIM reseat:** After power-on, `mmcli -L` returned no modems and `lsusb | grep 1199` returned nothing, at 8 minutes of uptime.
7. **USB topology check:** `lsusb -t` and `dmesg` showed hub 1-4 with FTDI on port 2 and Teensy on port 3 enumerating normally. Port 1 had no device and no error messages.
8. **Full power cut:** Power disconnected for 30 seconds and the modem seating checked. After boot the result was identical: port 1 empty and no modems found.
9. **Restoration and decision to stop:** The connectivity cron job was reactivated. No firmware flashing was attempted, since the modem is not reachable.

CONCLUSION AND RECOMMENDATION

The modem of N162AV is no longer present on the USB bus. It was working at the start of the session and disappeared after the unit was powered off, and it did not return after a full power cut. The other devices on the same hub enumerate normally, which points to the modem or its connection and not to the hub. The original `sim-missing` symptom is unresolved.

Recommended action: Issue a maintenance work order for physical inspection of the modem module, its USB connector and cable, and the corresponding hub port, with replacement of the modem as the likely fix. If the SIM is to be reused, test it in a known-good unit first. Do not attempt firmware flashing.

APPENDIX: KEY COMMAND OUTPUT

```
[AVA:N162AV] # mmcli -m 0 (start of session)
state: failed | failed reason: sim-missing | power state: on | signal quality: 0% (cached)
# qmicli -d /dev/cdc-wdm0 --uim-get-card-status
error: couldn't create client for the 'uim' service: CID allocation failed in the CTL client: endpoint hangup
# pgrep -c pppd
8
After the logical reset, SIM reseat power-off and power-on (8 min uptime):
```

```
# mmcli -L
No modems were found
# lsusb | grep -i 1199
(no output)
# lsusb -t (excerpt)
|__ Port 4: Dev 4, If 0, Class=Hub, Driver=hub/4p, 480M
|__ Port 3: Dev 6, If 0, Class=Communications, Driver=cdc_acm, 480M
|__ Port 2: Dev 5, If 0, Class=Vendor Specific Class, Driver=ftdi_sio, 12M
(no device on hub Port 1)
# dmesg | grep -E "usb 1-4|enumerate|Cannot enable" (excerpt)
[ 5.181727] usb 1-4: new high-speed USB device number 4 using xhci_hcd
[ 5.653703] usb 1-4.2: new full-speed USB device number 5 using xhci_hcd
[ 5.885731] usb 1-4.3: new high-speed USB device number 6 using xhci_hcd
(no entries for usb 1-4.1 and no enumeration errors)
After the full power cut:
# lsusb -t -> same topology, hub Port 1 empty
# mmcli -L -> No modems were found
```